

Designing Effective Visualizations for Geographic Data

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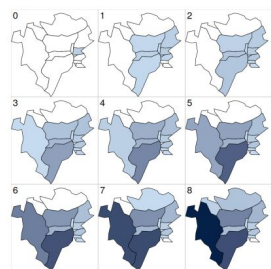


Research Interests

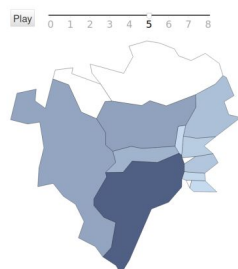
- **Geo-temporal** data
- Data with **uncertainty**
- **Sustainability** related projects



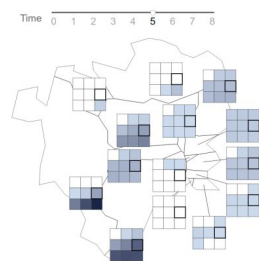
(i) Perception studies



(a) Small-multiple Maps

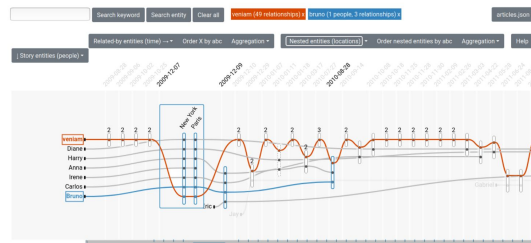


(b) Animated Map



(c) Glyphs over Map

(ii) Hypergraphs



Time: 2009-12-07

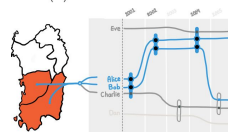
People: Anna, Irene, Harry, Diane, Carlos, Bruno

Locations: New York, Paris

Source: cb7579.json

"Népal et ex non. Surtout qui est exceptionnel Paris Bruno commode. Vingt sont allé magna, Anna non eu Carlos que Diane suite nulle from. Data ex temps descend et deux volaplate présent. Lorem sint New York small ultramc representant velum error. Ex exception amet 2009-12-07 et Irene ex sunt volaplate exception. Lorem data ex adipiscing. Parietur aliquip aliquip sint enim. Henry non duis nulli et cupiditate."

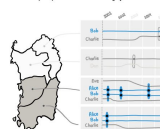
(A) Coordinated views



(B) Map Glyphs



(C) Time Glyphs



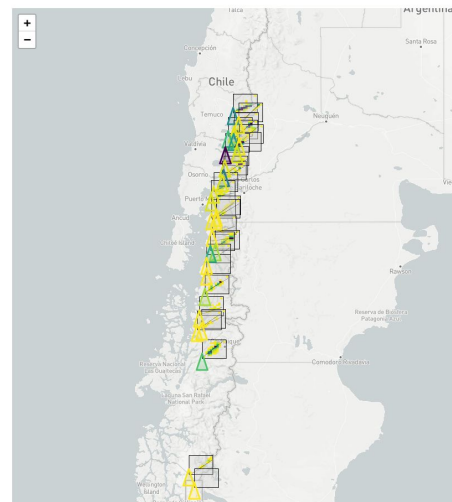
(iii) Uncertainty

Volcanic Centers

Sample Number ☐ Deactivate colormap on volcano

Distance between a scatter point and regression line: ☐ Deactivate sample pointers on sample

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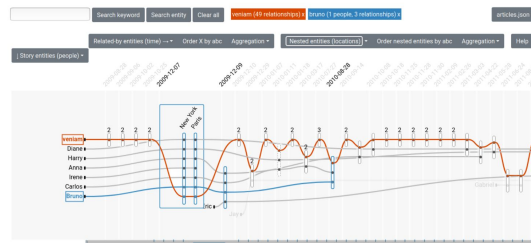


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(c) Glyphs over Map

(ii) Hypergraphs



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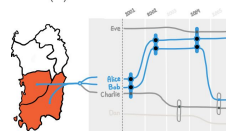
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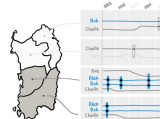
(A) Coordinated views



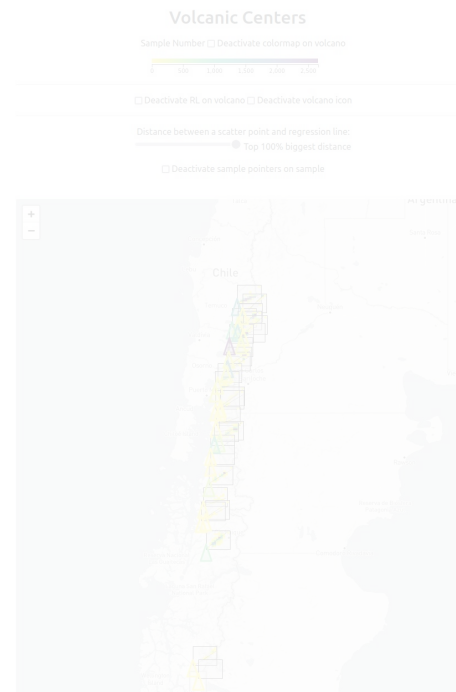
(B) Map Glyphs



(C) Time Glyphs



(iii) Uncertainty



Relationships of entities mentioned in news articles

Collab with:



Relationships of entities mentioned in news articles

Collab with:



"Laboris do sint aliqua eu magna aliqua. Ullamco sunt ex est **Accra** consequat magna. Adipiscing cillum ad eiusmod aliquip esse. Excepteur ipsum officia voluptate laboris. Labore sed ea anim do nulla. Quis dolore **Gabriel** magna nisi, amet **2011-12-20** id fugiat nisi culpa ullamco, irure aliquip non labore. Anim deserunt **Fanny** do quis, sit consequat eiusmod non **Lima** nisi enim sunt aliqua. Quis deserunt consequat aliquip, incididunt voluptate commodo mollit do anim, eiusmod dolore officia cupidatat exercitation. Sit non cupidatat eu ut ipsum laboris. Pariatur mollit deserunt consectetur velit irure quis."

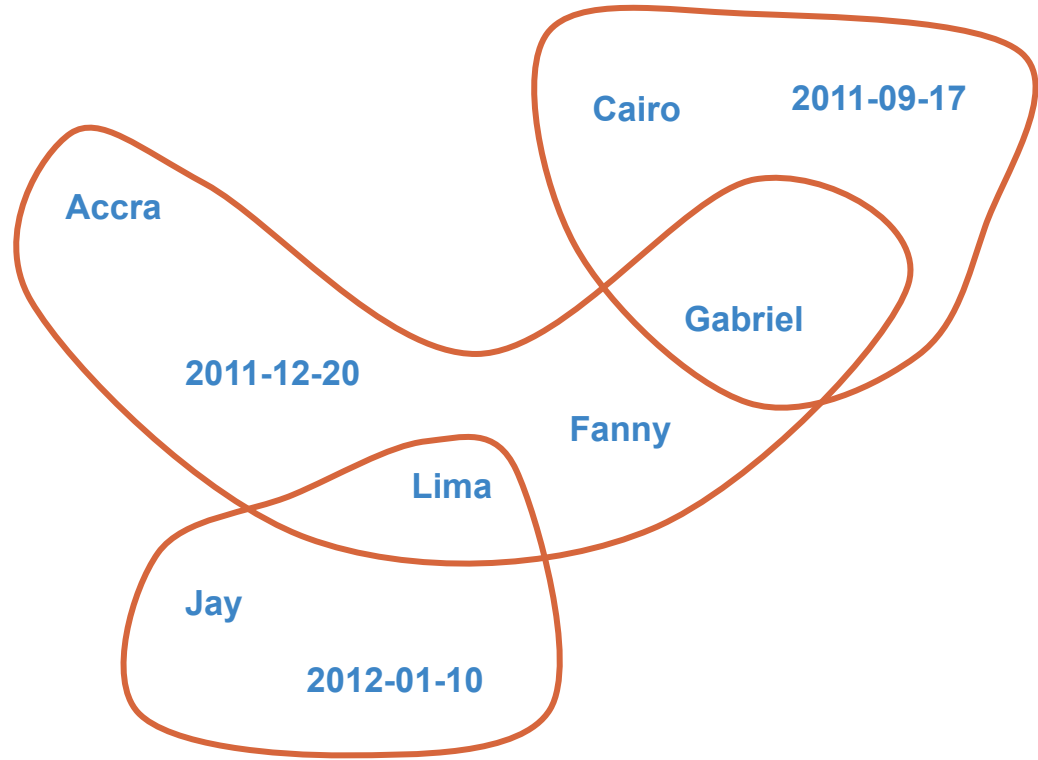
Relationships of entities mentioned in news articles

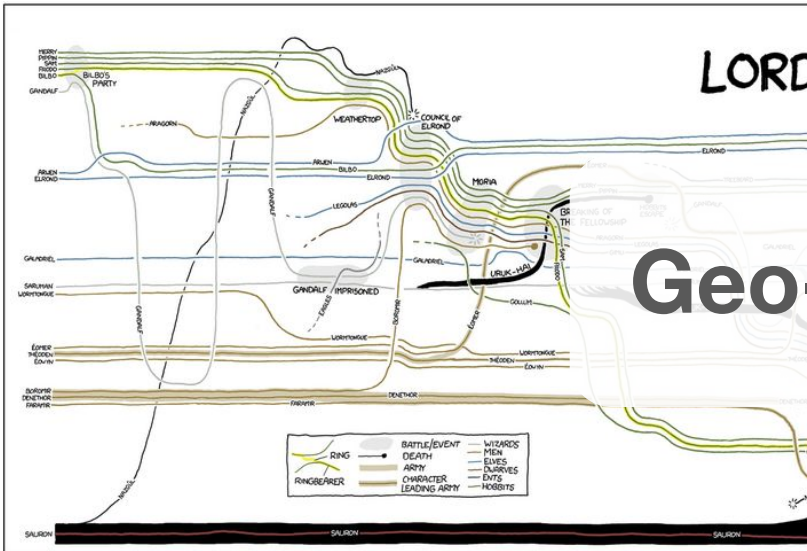
Collab with:



Hypergraphs

- Visualizing these datasets is an emerging research field
- Other domains: co-authorship, social networks, historical analysis, etc.





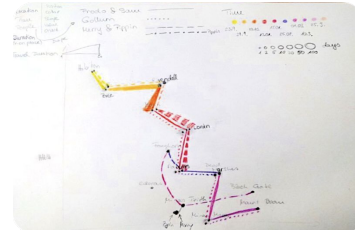
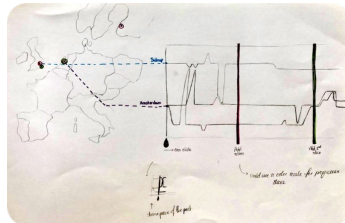
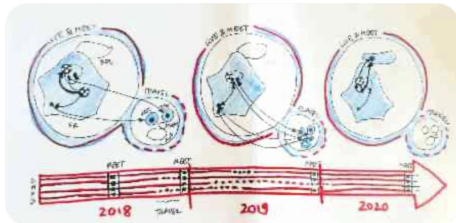
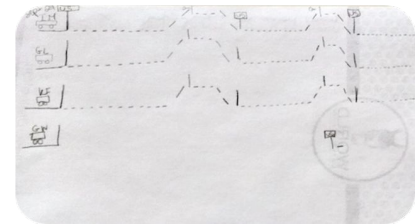
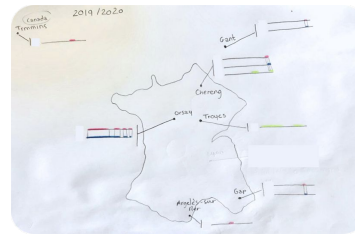
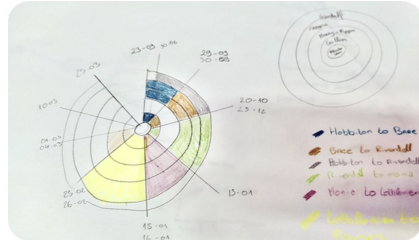
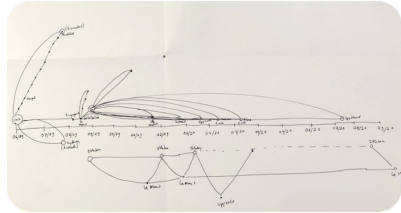
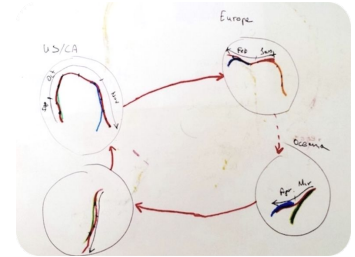
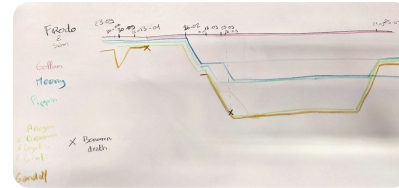
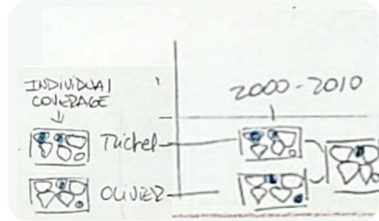
Source: <https://xkcd.com/657/>

Geo-Storylines

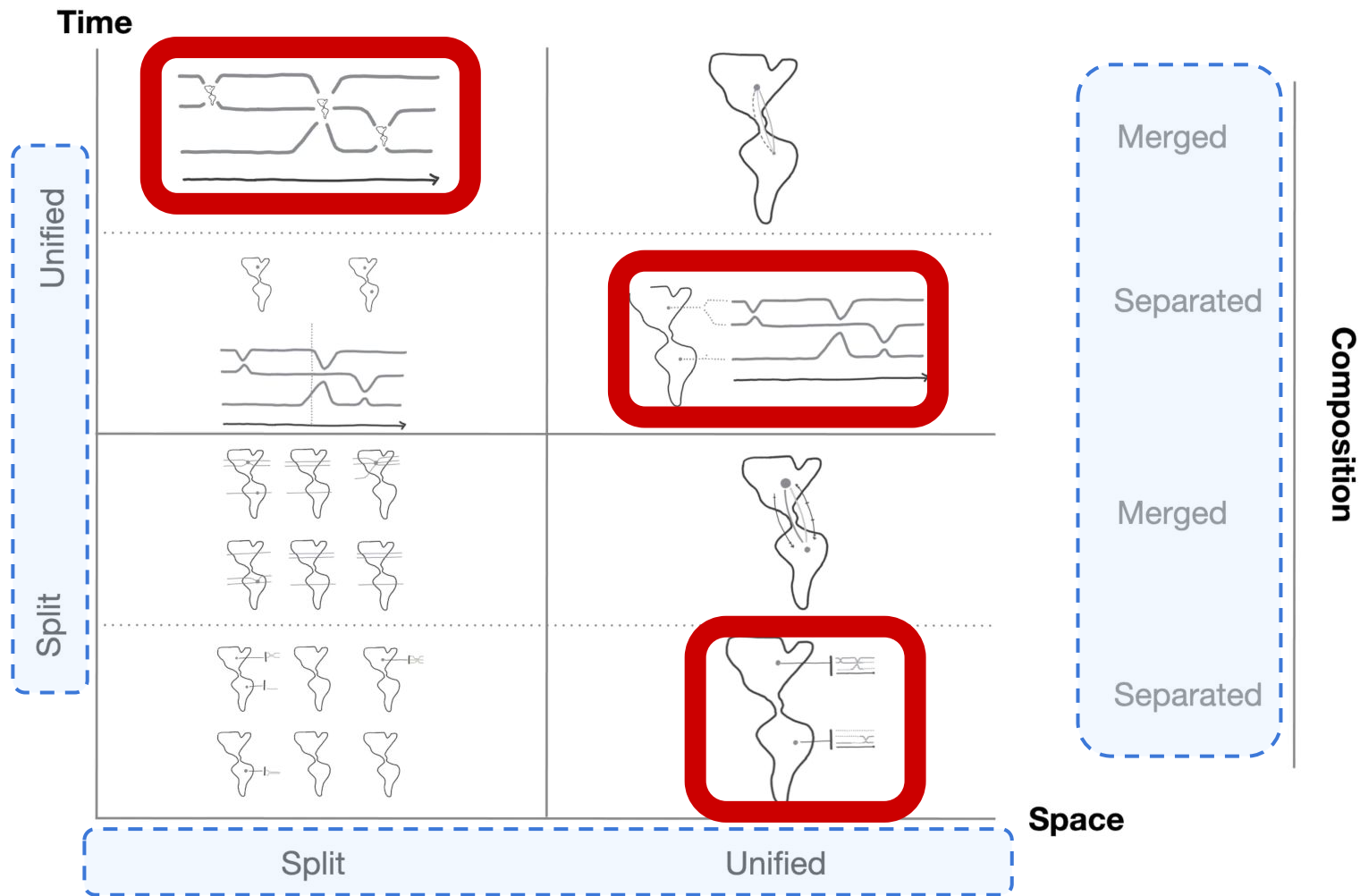


Source: lotrproject.com

We conducted generative workshops



67
sketches



Demo

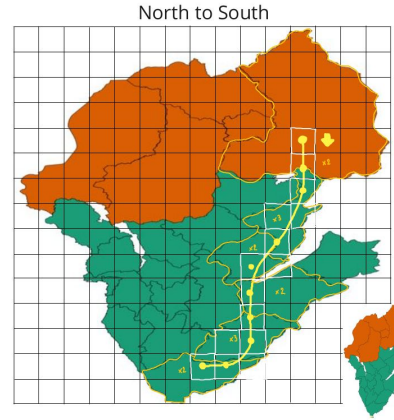
<https://geostorylines.lisn.upsaclay.fr/>

Two evaluations

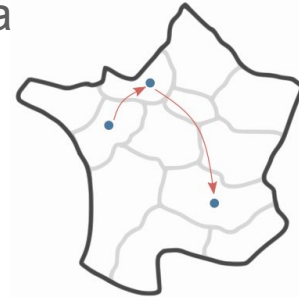


Icon created by Flat Icons

1) User study with 18 participants

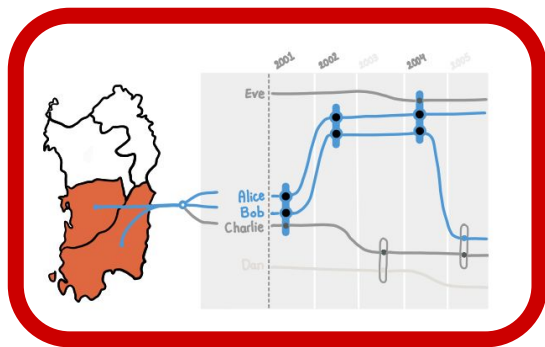


2) Domain experts feedback using their own data



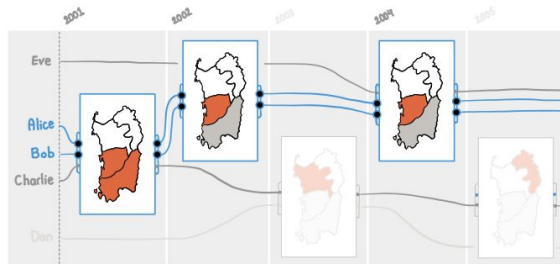
Overall Results

(A) Coordinated Views



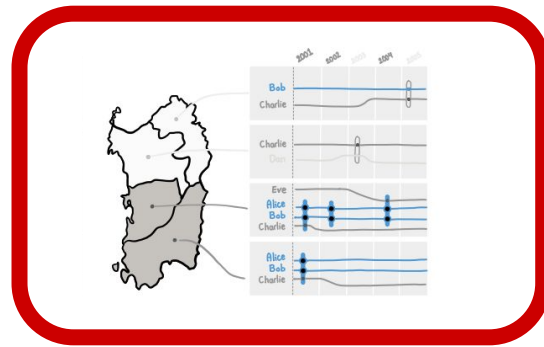
Best Overall

(B) Map Glyphs



No errors searching
locations

(C) Time Glyphs



Worst overall

Experts: (A) and (C) are Complementary !!!

(i) Perception studies



(a) Small-multiple Maps

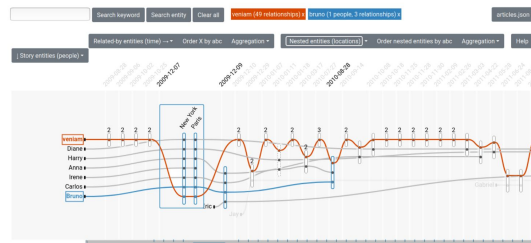


(b) Animated Map



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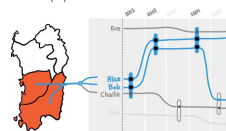
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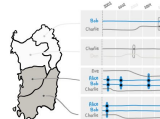
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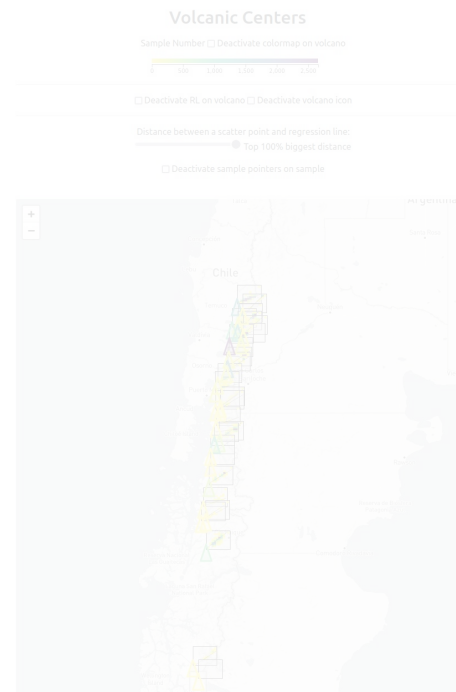
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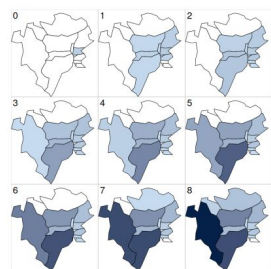
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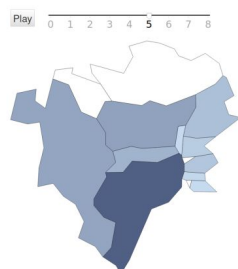
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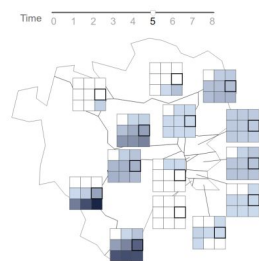
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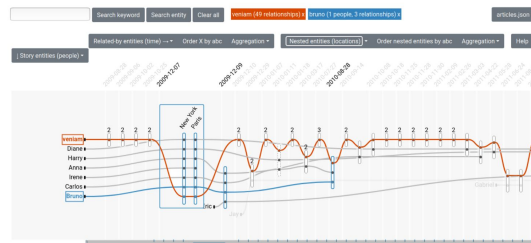


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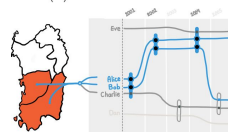
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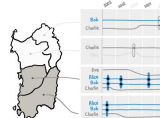
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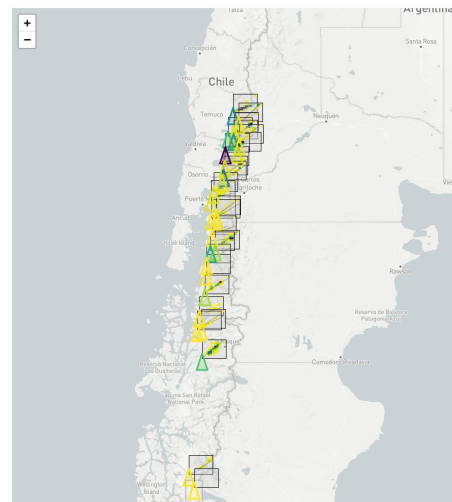
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(i)

Perception studies

Anastasia Bezerianos, Emmanuel
Pietriga

[CHI'19, IEEEVIS'20]

(ii)

Hypergraphs

Anastasia Bezerianos, Golina
Hulstein, Emmanuel Pietriga

[IV'21, IEEEVIS'22]

(iii)

Uncertainty

Anastasia Bezerianos, Guillaume
Delpech, Consuelo Martinez
Fontaine

[ACM CHI'25, to appear]



Research Interests

- **Geo** data
- Data with **uncertainty**
- **Sustainability** related projects

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